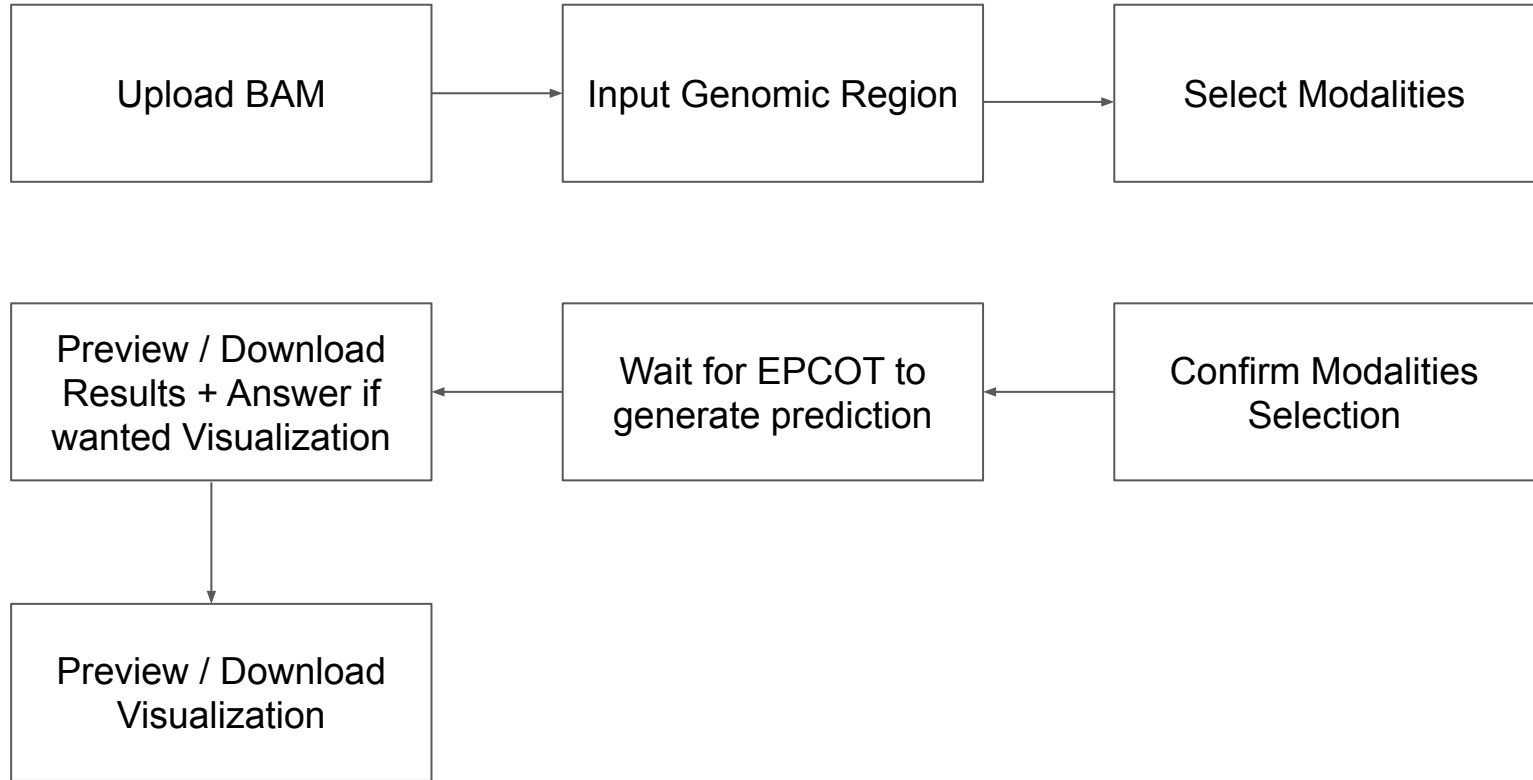


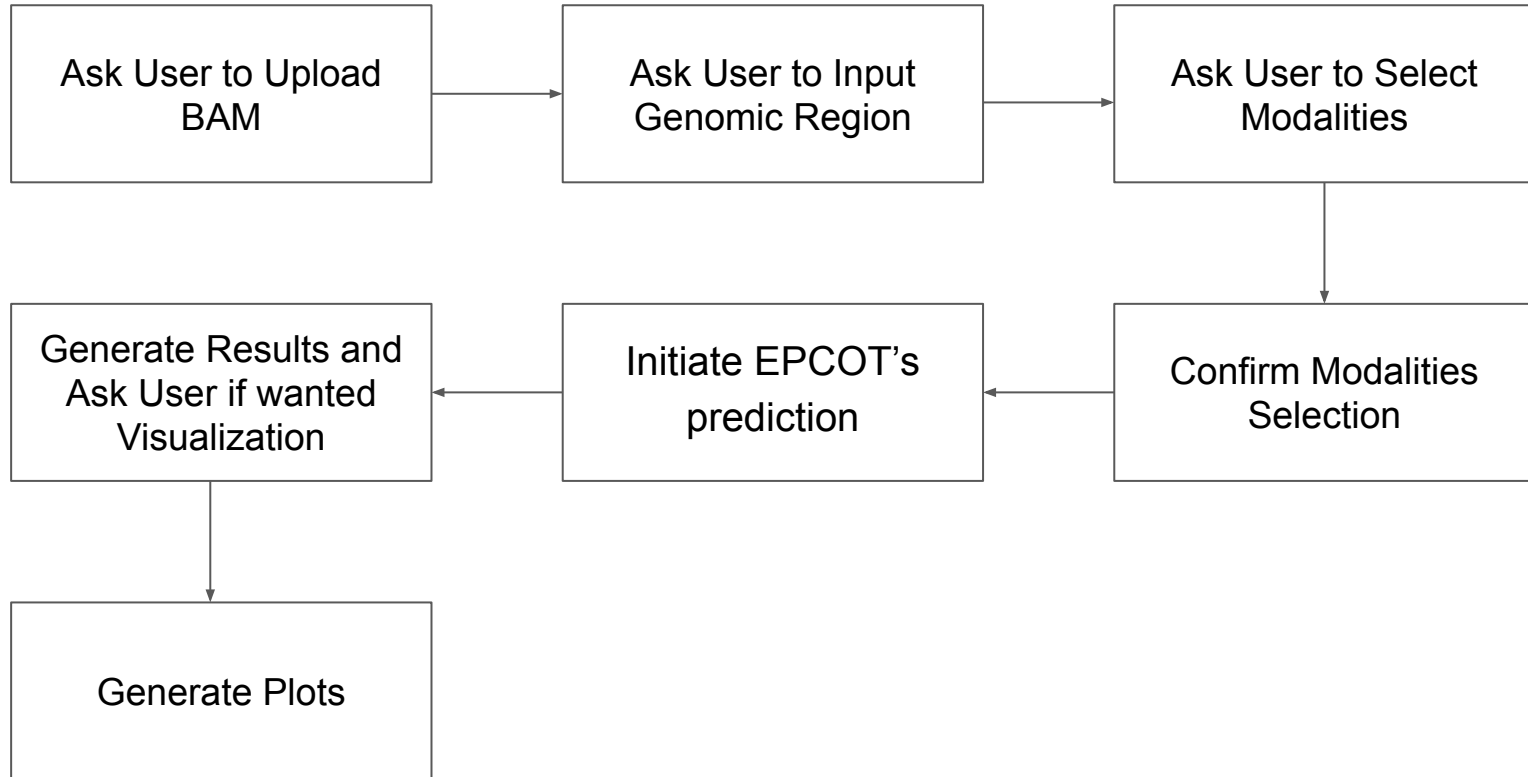
EPCOT Agent v 2.0

- 1) EPCOT Agent 2.0 Detailed Introduction
- 2) EPCOT Agent 3.0 Next Steps

EPCOT Agent v 2.0: Workflow for the User



EPCOT Agent v 2.0: Workflow for Agent



EPCOT Agent v 2.0: Supported Modalities

```
import pickle
with open('spleen.pickle', 'rb') as file:
    spleen_pickle = pickle.load(file)
## All predicted modalities
spleen_pickle['epi']
spleen_pickle['rna'] # expected shape: (11123, 500, 3)
spleen_pickle['bru'] # expected shape: (11123, 500, 3)
spleen_pickle['microc'] # expected shape: (11123, 500, 500, 2)

spleen_pickle['intacthic'] # expected shape: (11123, 500, 2)
spleen_pickle['rna_strand'] # expected shape: (11123, 500, 2)
spleen_pickle['external_tf']
spleen_pickle['tt'] # expected shape: (11123, 500, 2)
spleen_pickle['groseq'] # expected shape: (11123, 500, 2)
spleen_pickle['grocap'] # expected shape: (11123, 500, 4)
spleen_pickle['proseq'] # expected shape: (11123, 500, 3)
spleen_pickle['netcage'] # expected shape: (11123, 500, 2)
spleen_pickle['starr'] # expected shape: (11123, 500, 1)
```

EPCOT Agent v 2.0: Supported Genomic Region

- 1) Chromosome number: 1-22, X
- 2) Start position: any number $\geq$ EPCOT window's smallest position for that chromosome number
- 3) End position: any number $\leq$ EPCOT window's largest position for that chromosome number
 - Predicting region has to be $\leq$ 600 kb (i.e., `end pos` - `start pos` $\leq$ 600000)
 - Predicting region will be matched with EPCOT window's overlapped region
 - a) If the requested region exactly matched with one region from EPCOT window, run the requested region
 - b) If the requested region didn't match exactly with one region from EPCOT window, run the region overlap with the requested region (i.e., $\text{window_end} > \text{start pos}$ AND $\text{window_start} < \text{end pos}$)

EPCOT window file:

/nfs/turbo/umms-drjeliu/usr/luosanj/EPCOTv2_gradio/EPCOTv2/erna/input_region_600k
b.bed

EPCOT Agent v 2.0: Detailed Agent Workflow

- 1) Prompt the user to upload a chromatin accessibility file (.bam).
(Any file with a .bam extension is accepted now, as a preprocessed .pkl file is used in the backend.)
- 2) Request the user to input the genomic region for prediction, including chromosome number, start position, and end position.
- 3) Request the user to specify the modality or modalities to be predicted.
- 4) Confirm the selected modalities with the user before proceeding.
- 5) Initiate EPCOT's prediction process upon user confirmation.
- 6) Offer user the option to generate visualization plots of the prediction results, with capabilities to download them via the `/chat` response.

EPCOT Agent v 3.0: Detailed Agent Workflow

- 1) Prompt the user to upload a chromatin accessibility file (.bam) and convert it into .pkl file in the backend
- 2) Request the user to input the genomic region for prediction, including chromosome number, start position, and end position.
- 3) Request the user to specify the modality or modalities to be predicted.
- 4) Confirm the selected modalities with the user before proceeding.
- 5) Initiate EPCOT's prediction process upon user confirmation.
- 6) Offer user the option to generate visualization plots of the prediction results, with capabilities to preview and download outputs.

EPCOT Agent v 3.0: Supported Modalities

```
import pickle
with open('spleen.pickle', 'rb') as file:
    spleen_pickle = pickle.load(file)
## All predicted modalities
spleen_pickle['epi']
spleen_pickle['rna'] # expected shape: (11123, 500, 3)
spleen_pickle['bru'] # expected shape: (11123, 500, 3)
spleen_pickle['microc'] # expected shape: (11123, 500, 500, 2)
spleen_pickle['hic'] # expected shape: (11123, 100, 100, 3) Hi-C is predicted at 5kb resolution
spleen_pickle['intacthic'] # expected shape: (11123, 500, 2)
spleen_pickle['rna_strand'] # expected shape: (11123, 500, 2)
spleen_pickle['external_tf']
spleen_pickle['tt'] # expected shape: (11123, 500, 2)
spleen_pickle['groseq'] # expected shape: (11123, 500, 2)
spleen_pickle['grocap'] # expected shape: (11123, 500, 4)
spleen_pickle['proseq'] # expected shape: (11123, 500, 3)
spleen_pickle['netcage'] # expected shape: (11123, 500, 2)
spleen_pickle['starr'] # expected shape: (11123, 500, 1)
```